Yifei Chen

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Technology of CSIRO, Melbourne, Australia

Profile

Robotics and reinforcement learning researcher at CSIRO with a PhD in Artificial Intelligence. My research examines how agents learn from demonstrations and corrective feedback, and how their decisions generalize when observations or training conditions change. I develop and evaluate learning methods in simulation and have experience testing policies on physical robots.

Technical Skills

Research areas: Reinforcement learning, imitation learning, reward shaping, robot learning, generalization and robustness
Methods and tools: Python, PyTorch, TensorFlow; MuJoCo, CoppeliaSim; robotic manipulation, simulation experiments, policy evaluation and physical robot deployment

Professional Experience

Technology of CSIRO, Australia Jan. 2024 - present
Early Research Career Fellow
Quantitative Imaging Team (Jan. 2024 - Mar. 2026), Manager: Xun Li
Human-robot interaction team (Mar. 2026 - present), Manager: Leimin Tian

- Investigated generalization in action chunking transformer (ACT), developing and evaluating methods to reduce reliance on spurious observation features across in-distribution and out-of-distribution settings.
- Designed controlled offline-to-online reinforcement learning experiments to study when expert interventions help or hinder learning, comparing intervention policies, reward definitions, and transition-retention strategies.
- Built training and evaluation workflows for robot learning in MuJoCo and established manipulation benchmarks.
- Collected demonstrations and deployed policies on the ROS2-based physical ALOHA arms, investigating camera configuration and execution stability.

University of New South Wales, Australia Jan. 2023 - Dec. 2023
Visiting Researcher. Supervisor: Dr. Francisco Cruz Naranjo

- Developed APF-DDPG by integrating our proposed adaptive potential function (APF) with Deep Deterministic Policy Gradient (DDPG) for continuous robotic reaching tasks.
- Trained and evaluated Baxter right-arm reaching policies in CoppeliaSim across 20 runs, improving learning performance and consistency over the DDPG baseline.
- Deployed the trained policy on a physical Baxter robot and evaluated it on the same reaching task used in CoppeliaSim.

Education

University of Groningen (RUG), Groningen, the Netherlands Oct. 2018 - Apr. 2025
Ph.D. in Artificial Intelligence
Thesis title: Beyond Value Estimation: Adaptive Potential Functions For Reinforcement Learning
Supervisors: Prof. dr. Lambert Schomaker and Dr. Marco Wiering

Zhejiang University, Hangzhou, China Sep. 2015 - Jun. 2018
M.Sc. in Underwater Wireless Optical Communication
Award: National Scholarship for Graduate Students; Excellent Postgraduate Students' Award
Thesis title: Experimental Research on Air-water Wireless Optical Communication System
Supervisor: Prof. Jing Xu

Selected Publications and Preprints

When Do Interventions Help Q-Chunking? An Empirical Study on Pen Manipulation. NeurIPS Workshop
Yifei Chen*, Dana Kulic, and Leimin Tian.
Accepted at the NeurIPS 2026 EvoRobust Workshop

Improving Generalization Ability of Robotic Imitation Learning by Resolving Causal Confusion in Observations. Preprint
Yifei Chen*, Yuzhe Zhang, Giovanni Durso, Nicholas Lawrance, and Brendan Tidd.
arXiv preprint 2025

Boosting Reinforcement Learning Algorithms in Continuous Robotic Reaching Tasks using Adaptive Potential Functions. AJCAI 2024
Yifei Chen*, Lambert Schomaker, Francisco Cruz
The 37th Australasian Joint Conference on Artificial Intelligence

Reinforcement Learning with Potential Functions Trained to Discriminate Good and Bad States. IJCNN 2021
Yifei Chen*, Hamidreza Kasaei, Lambert Schomaker, Marco A. Wiering
International Joint Conference on Neural Networks

An Investigation Into the Effect of the Learning Rate on Overestimation Bias of Connectionist Q-learning. ICAART 2021
Yifei Chen*, Lambert Schomaker, Marco A. Wiering
The 13th International Conference on Agents and Artificial Intelligence

corresponding author*